

Question Paper

Exam Date & Time: 22-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Digital design verifications [ECE_4407]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Develop a step-by-step verification plan in System Verilog to test and validate the functionality of a 4-bit adder design. (5)
A) Course Outcome: "CO5" / Cognitive Level: "Application"
B) Implement a functional coverage model in SystemVerilog to ensure all possible input combinations are tested for an XOR gate. (3)
Course Outcome: "CO1" / Cognitive Level: "Understand"
C) List and describe some common verification techniques and tools used in VLSI design platforms. (2)
Course Outcome: "CO1" / Cognitive Level: "Remember"
- 2) Analyze the differences between functional verification and formal verification with suitable examples, highlighting their methodologies, advantages, and limitations. (5)
A) Course Outcome: "CO2" / Cognitive Level: "Analysis"
B) Analyze the steps involved in achieving verification closure for a complex SoC design, explaining how each step contributes to meeting functional and coverage goals. (3)
Course Outcome: "CO2" / Cognitive Level: "Analysis"
C) Explain strategies used in logic verification and its role to ensure design correctness. (2)
Course Outcome: "CO2" / Cognitive Level: "Understand"
- 3) Compute System Verilog program for basic logic gates (AND, OR, XOR, XNOR) using assign statements and test bench program with input combination (Stimulus) of (00, 01, 10, 11) applied to validate the functionality of design under test (DUT). (5)
A) Course Outcome: "CO4" / Cognitive Level: "Application"
B) Classify and illustrate various data types in System-Verilog with examples to demonstrate their usage in hardware modeling and verification. (3)
Course Outcome: "CO3" / Cognitive Level: "Application"
C) Apply SystemVerilog assertions in a testbench to verify specific design behaviors and discuss its (2)

role in detecting protocol or timing violations.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) What are the key components of a testbench? Implement their functions using a simple DUT example (e.g., AND gate). (5)

A)

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Apply your understanding of verification methods to explain how formal verification is used in digital design and illustrate its advantages over simulation with suitable examples. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- C) Explain the main phases of a UVM testbench and describe the purpose of each phase in the verification process. (2)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Analyze the building blocks of UVM (such as sequence, driver, monitor, and scoreboard) and explain how it interacts to achieve functional coverage and reusability in a verification environment. (5)

A)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) Analyze Verilog, VHDL, and System-Verilog with respect to verification capabilities, data types, and modeling styles. (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- C) Define verification and its roles in VLSI design. (2)

Course Outcome: "CO1" / Cognitive Level: "Remember"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 ECE_4407:Digital design verifications

#	Course Outcome	Description	Marks
1	1	CO 1	12
2	2	CO 2	10
3	3	CO 3	5
4	4	CO 4	12
5	5	CO 5	11
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 ECE_4407:Digital design verifications

#	Level	No of Questions	Marks (%)
1	Remember	2	4(8%)
2	Understand	2	5(10%)
3	Application	7	25(50%)
4	Analysis	4	16(32%)
5	Create	0	0(0%)

6	Evaluate	0	0(0%)
	Total	15	50 (100%)